

Synthesizing Climate Risk: Hazards, Impacts, and Mitigation

An Analysis of IPCC AR6 WG1, WG2, and WG3 Data

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Course: AtmSci7072 Global Climate Change

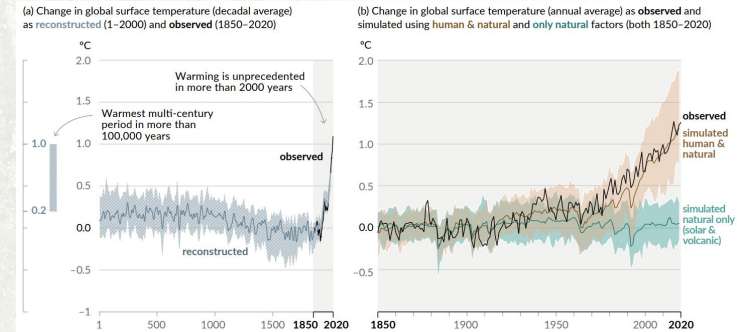
SID: 61444026S NTNU

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PHYSICAL SCIENCE

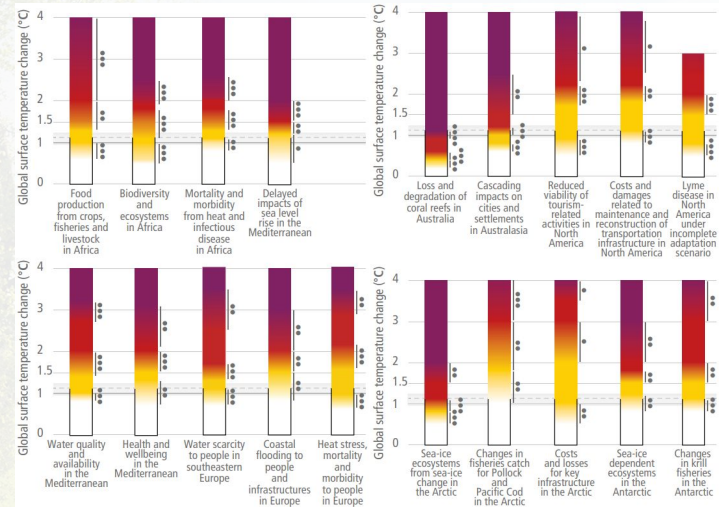
- The figure shows the fundamental physical hazard, **anthropogenic warming**, that drives the cascading risks and necessitates the policy responses detailed in the subsequent reports.
- Global surface temperatures are higher than they have been in at least **100,000 years**. Climate models CMIP6 also show that **human activity** has driven this fast warming over the past **170 years**.
- **Reflection:** I am surprised by how quickly modern temperatures have broken away from long term historical records. In the future, I want to study which regional temperature changes might cause these global models to speed up more than we expect.



History of global temperature change and causes of recent warming. (IPCC AR6 WG1, Figure SPM.1)

IMPACTS

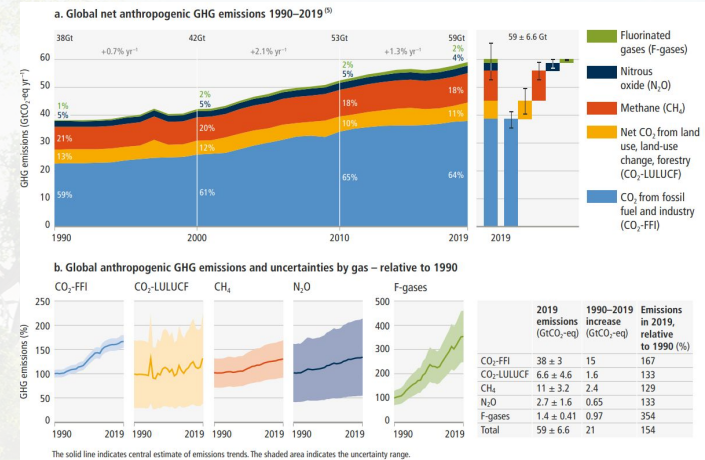
- The figure connects the physical temperature hazard (WG1) directly to local effects. It shows how global warming creates specific **regional vulnerability and exposure**.
- Even small increases in global warming cause severe risks to water security, food production, and ecosystems. These risks mostly hurt **poor communities** because they have the **lowest adaptive capacity**.
- **Reflection:** I think it is worried that how fast vulnerable human systems move from moderate to very high risk. I am curious to explore how we can better measure and fund local adaptive capacity to stop these high risk situations.



How climate risks for natural and human systems increase from 0°C to 5°C of global warming. The colors show the risk level, changing from white (undetectable) to purple (very high). (IPCC AR6 WG2, Figure SPM.3)

MITIGATION CHALLENGE

- The figure connects to WG1 and WG2 by showing the main human problem: **rising greenhouse gas emissions**, which increase the physical hazard and make regional impacts much worse.
- The average annual rate of growth slowed from 2.1% per year in 2000-2009 to 1.3% per year in 2010-2019. In 2019, global human caused emissions reached **59 GtCO₂-eq**, higher than any previous decade.
- **Reflection:** Because renewable energy has become much cheaper in the last decade, it surprises me that total emissions are still growing. I am curious to how developing countries can skip fossil fuel use by adopting these cheaper technologies.



Global net anthropogenic GHG emissions (GtCO₂-eq yr⁻¹) 1990–2019. Even the annual growth rate slowed down recently, total human caused GHG emissions continued to rise, reaching a record 59 GtCO₂-eq in 2019. (*IPCC AR6 WG3, Figure SPM.3*)

SYNTHESIS AND SELF STUDY

- Feedback Loop: Emissions (WG3) → Hazards (WG1) → Risks to vulnerable regions (WG2)
- Least Developed Countries (LDCs) and Small Island Developing States (SIDS) have much lower per capita emissions than the global average yet face the highest risks.
→ Effective climate policy cannot rely on mitigation or adaptation alone, it must be a strategy that reduces the emissions while simultaneously lowering localized vulnerability.
- Reflections: I am surprised by the rapid technological progress, from 2010 to 2019, there have been rapid cost drops for solar (-85%), wind (-55%), and batteries (-85%). And it is inspiring that at least 18 countries have sustained GHG emission reductions for longer than 10 years.
- Future questions to study: How can we accelerate the transfer of these low emission technologies to developing countries, where adoption currently lags due to weaker enabling conditions?

THANK YOU!

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